



OPEN A network meta-analysis of efficacy and safety of adjuvant targeted therapy or immunotherapy in non-small cell lung cancer

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The integration of targeted therapy and immunotherapy into adjuvant treatments for early-stage non-small cell lung cancer (NSCLC) remains challenging. This study aimed to compare the efficacy and safety of all available adjuvant treatment options. Randomized controlled trials (RCTs) published up to August 15, 2023, were identified from MEDLINE, Scopus, and Cochrane CENTRAL. RCTs were included if they studied early-stage NSCLC and compared any adjuvant systemic targeted therapy or immunotherapy with adjuvant chemotherapy/placebo. This study was registered with PROSPERO, CRD42022351290. Individual patient data were generated based on data extracted from Kaplan–Meier curves. A parametric survival model was used to estimate the median time for disease-free survival (DFS) and overall survival (OS). A two-stage network meta-analysis (NMA) was applied to estimate the hazard ratio (HR) for DFS and OS, in addition to the risk ratio (RR) of severe adverse events (SAE). Nineteen RCTs ($n=9,438$) were included for assessing the effects of epidermal growth factor receptor tyrosine kinase inhibitors (EGFR-TKIs), vascular endothelial growth factor (VEGF) inhibitors, immune checkpoint inhibitor (ICI) immunotherapy, and non-ICI immunotherapy. DFS in EGFR-TKIs and ICI immunotherapy was longer than chemotherapy/placebo, with a median time of 69.24, 53.47, and 39.49 months, respectively. EGFR-TKIs had a 46% significantly lower risk of recurrence [HR = 0.54 (95% CI: 0.38, 0.77)] than chemotherapy/placebo. Both EGFR-TKIs and ICI immunotherapy appeared to improve OS compared to chemotherapy or placebo, but both treatments also had an increased risk of SAE; however, neither result was statistically significant. This study indicates that EGFR-TKIs might be the best treatment regimen for reducing recurrence with an intermediate risk of SAE.

Keywords Non-small cell lung cancer, Adjuvant therapy, Targeted therapy, Immunotherapy, Network meta-analysis, Disease-free survival

Abbreviations

NSCLC	Non-small cell lung cancer
RCT	Randomized controlled trials
HR	Hazard ratio
DFS	Disease-free survival
OS	Overall survival
NMA	Network meta-analysis
RR	Risk ratio
SAE	Severe adverse events
EGFR-TKIs	Epidermal growth factor receptor tyrosine kinase inhibitors
VEGF	Vascular endothelial growth factor
ICI	Immune checkpoint inhibitor
SR	Systematic reviews

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MAs	Meta-analyses
AIC	Akaike's information criterion
CI	Confidence interval
SUCRA	Surface under the cumulative ranking curve
CINeMA	Confidence in network meta-analysis

Lung cancer is the most frequently diagnosed malignancy worldwide, with an estimated 2.2 million new cases in 2020¹. The 5-year survival rate across all stages has been reported to be as low as 23% to 25%². Early-stage non-small cell lung cancer (NSCLC), stage I to IIIA disease, accounted for approximately 50% of all diagnoses³. The main treatment for this stage is surgical resection, with the addition of adjuvant chemotherapy providing an absolute 5-year survival benefit of approximately 5%, compared with surgery alone^{4,5}.

Targeted therapy and immunotherapy are increasingly used to lower disease recurrence in early-stage NSCLC. For instance, osimertinib, the third-generation epidermal growth factor receptor tyrosine kinase inhibitor (EGFR-TKI), improved disease-free survival (DFS) when added to adjuvant chemotherapy⁶. In addition, recent randomized controlled trials (RCTs) showed that immune checkpoint inhibitors (ICI) (e.g., atezolizumab and pembrolizumab) significantly improved DFS in early-stage NSCLC patients when added to adjuvant chemotherapy^{7,8}. Therefore, it remains challenging to identify the optimal adjuvant systemic treatment regimen for early-stage NSCLC.

Several systematic reviews (SR) and meta-analyses (MAs)^{9–13} have assessed the efficacy of adjuvant systemic treatments in early-stage NSCLC. However, these SRMAs have focused on EGFR-TKIs compared to chemotherapy or placebo. Three network meta-analyses (NMAs)^{14–16} recently evaluated the efficacy and safety of adjuvant EGFR-TKIs with or without chemotherapy in early-stage NSCLC. They found that adjuvant EGFR-TKIs, specifically osimertinib, provided the longest DFS benefits, while adding EGFR-TKIs to chemotherapy caused the most toxic effects compared with chemotherapy alone. However, these reviews did not include all available treatment options, such as vascular endothelial growth factor (VEGF) inhibitors or novel immunotherapies, and they primarily focused on relative treatment effects measured by a hazard ratio (HR) without providing the median time of DFS and overall survival (OS), which is essential information for effective treatment planning.

None of the NMAs have yet to fully include all available treatment regimens for resected early-stage NSCLC patients. This SRNMA was therefore conducted to assess all current adjuvant systemic treatment regimens for early-stage NSCLC, including targeted therapy and immunotherapy, to prolong DFS and/or OS and lower the incidence of severe adverse events (SAE).

Methods

Protocol registration and reporting

The SRNMA was performed according to the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) extension for NMAs¹⁷, and its protocol was registered in PROSPERO (CRD42022351290).

Eligibility criteria

RCTs were included if they met the following criteria: studied in patients with completely resected early-stage (stage I – IIIA) NSCLCs; compared any adjuvant systemic targeted therapy (i.e., EGFR-TKIs, VEGF inhibitor) or immunotherapy (i.e., ICI immunotherapy, non-ICI immunotherapy) with adjuvant chemotherapy alone or any other systemic interventions; and reported at least one of the following outcomes, including DFS, OS, or SAE. If multiple publications of the same study were detected, the most recent and/or complete reports were selected. RCTs were excluded if they had insufficient data for pooling after three contact attempts with authors every two weeks or if published in a language unable to be translated by Google Translate.

Search strategy

The relevant studies were identified from MEDLINE via PubMed, Scopus, Cochrane Central Register of Controlled Trials, and conference proceedings from major medical oncology meetings through August 15, 2023. The references of selected studies and those included in previous SRs were also reviewed to identify potentially relevant studies. The search terms and search strategies were constructed based on patients (early-stage NSCLC), interventions (adjuvant therapy, EGFR-TKIs, VEGF inhibitors, immunotherapy), outcomes of interest (DFS, OS, SAE), and study design (RCT). The search terms and strategies are presented in supporting information (Supplementary Table S1).

Study selection

Titles and abstracts of identified studies were independently selected by two reviewers (S.S. and P.P.) based on the eligibility criteria. Full articles were subsequently reviewed if a decision could not be made at the initial step. Any disagreements were discussed and resolved with the author team.

Interventions

Interventions of interest were EGFR-TKIs targeted therapy (i.e., gefitinib, erlotinib, icotinib, afatinib, dacomitinib, and osimertinib), VEGF-inhibitors (i.e., bevacizumab), ICI immunotherapy (i.e., ipilimumab, nivolumab, pembrolizumab, atezolizumab, avelumab, and durvalumab), non-ICI immunotherapy (i.e., MAGE-A3 immunotherapeutic), and chemotherapy/placebo.

Outcomes of interest

The primary outcome was DFS, which was defined as the time from randomization to the first recurrence, distant metastasis, or death from any cause. The secondary outcomes were OS and SAE. OS was defined as the

time from randomization to death due to any cause. SAE included at least grade 3 or higher adverse events (i.e., life-threatening events, events requiring hospitalization) according to the National Cancer Institute's Common Terminology Criteria for Adverse Events (NCI-CTCAE)¹⁸.

Data extraction

Two reviewers (S.S. and P.P.) independently extracted data using a standardized data extraction form: study characteristics (i.e., author, year of publication, phase of study, region); patient characteristics (i.e., age, sex, race, Eastern Cooperative Oncology Group (ECOG) performance status, type of surgery, staging, histological type, adjuvant chemotherapy received, EGFR mutation status); intervention characteristics (i.e., medication name, classification, duration of treatment); and outcomes (i.e., DFS, OS, SAE). To pool treatment effects, adjusted hazard ratio (HR) with 95% confidence interval (CI) were extracted. Additionally, data were extracted from Kaplan–Meier curve using the WebPlotDigitizer (version 4.5)^{19,20}. Extracted data included probabilities of DFS/OS, number of events, and number of patients at risk at each distinct time point. These data were used to simulate individual patient time-to-event data²¹. The simulated data were validated by benchmarking against the original reports in term of the number of events, median survival times, and HRs, to ensure consistency with the original publications.

Risk of bias assessment

Two reviewers (S.S. and P.P.) independently assessed the quality of the studies. Any disagreements were resolved through discussions with the author team. A revised Cochrane risk-of-bias tool for randomized trials (RoB 2)²² was employed, encompassing 5 domains: randomization process, protocol deviations, missing outcome data, measurement of the outcome, and selection of the reported result. The overall risk of bias was categorized as “Low,” “Some concerns,” or “High.”

Statistical analysis

Direct meta-analysis (DMA) was performed when at least three studies were available. The median time, HRs, and risk ratios (RRs), along with 95% confidence intervals (CIs), were estimated and pooled across studies. A mixed-effect parametric survival model pooled the median time and HRs for DFS and OS; a random-effects DMA model pooled SAE if heterogeneity was present; otherwise, a fixed-effects model was used. Heterogeneity was assessed by Cochran's Q test and quantifying its degree by Higgins' I^2 statistic²³.

Meta-regression²⁴ was used to assess potential sources of heterogeneity by considering each covariate (e.g., EGFR mutation status, region, phase of RCT, time for follow-up, treatment duration, risk of bias, age, sex, race, ECOG performance status, type of surgery, staging, histological type, adjuvant chemotherapy received) one by one. Subgroup or sensitivity analysis was conducted accordingly for covariates that significantly reduced I^2 . Publication bias was assessed using Egger's test and a funnel plot²⁵; a contour-enhanced funnel plot was applied if asymmetry was detected²⁶.

For NMA, the treatment regimens were numerically coded for chemotherapy/placebo (1), EGFR-TKIs (2), VEGF inhibitors (3), ICI immunotherapy (4), and non-ICI immunotherapy (5).

For individual patient time-to-event data, a one-stage mixed-effect parametric survival model^{27,28} was constructed to estimate the median DFS/OS time by fitting treatments on time. Various time distributions were applied, including Weibull, exponential, log-logistic, log-normal, and generalized gamma. The model with the smallest value of Akaike's Information Criterion (AIC) was selected. The median time, along with the 95% confidence interval (CI), was then estimated.

For the relative treatment effects, a two-stage NMA was applied to estimate the HR for DFS and OS and RR for SAE across the network²⁹, and multiple relative treatment effects were subsequently estimated. The probability of being the best treatment, regarding a lower risk of recurrence or death and a lower incidence of SAE, was estimated and ranked using the rankogram and the surface under the cumulative ranking curve (SUCRA)³⁰ for all outcomes. Cluster ranking analysis^{29,30}, which considered both efficacy (DFS) and safety (SAE), was constructed to compare the risks and benefits of each treatment regimen. The assumption of transitivity was evaluated by comparing the co-variables across the studies. A design-by-treatment interaction model^{31,32} was used to evaluate the assumption of inconsistency. A comparison-adjusted funnel plot was used to assess publication bias. All analyses were performed using the STATA software package, version 17.0. Statistical significance was considered at a two-sided p-value of less than 0.05.

The certainty of evidence

The certainty of evidence estimated by the NMA was assessed using the framework proposed by the Confidence in Network Meta-Analysis (CINeMA) guideline^{33,34}. Six domains were evaluated: within-study bias, reporting bias, indirectness, imprecision, heterogeneity, and incoherence. Each relative treatment effect was finally judged as having a high, moderate, low, or very low level of evidence.

Results

Characteristics of included studies

Nineteen studies^{6–8,35–50}, comprising 9,438 patients, met the eligibility criteria and were included in this NMA, see Fig. 1. Characteristics of these 19 included studies are summarized in Table 1 and Supplementary Table S2. Most of the studies were conducted in the Asia-Pacific region. Ten studies (52.6%) were in EGFR mutation-positive NSCLC patients. The median age ranged from 55.0 to 66.5 years. Almost all patients had ECOG 0–1. The percentage of patients who underwent lobectomy ranged from 67.2% to 99.3%. The most common histological subtype was adenocarcinoma (76.8%). Approximately half of patients (56.5%) received adjuvant chemotherapy following surgical resection.

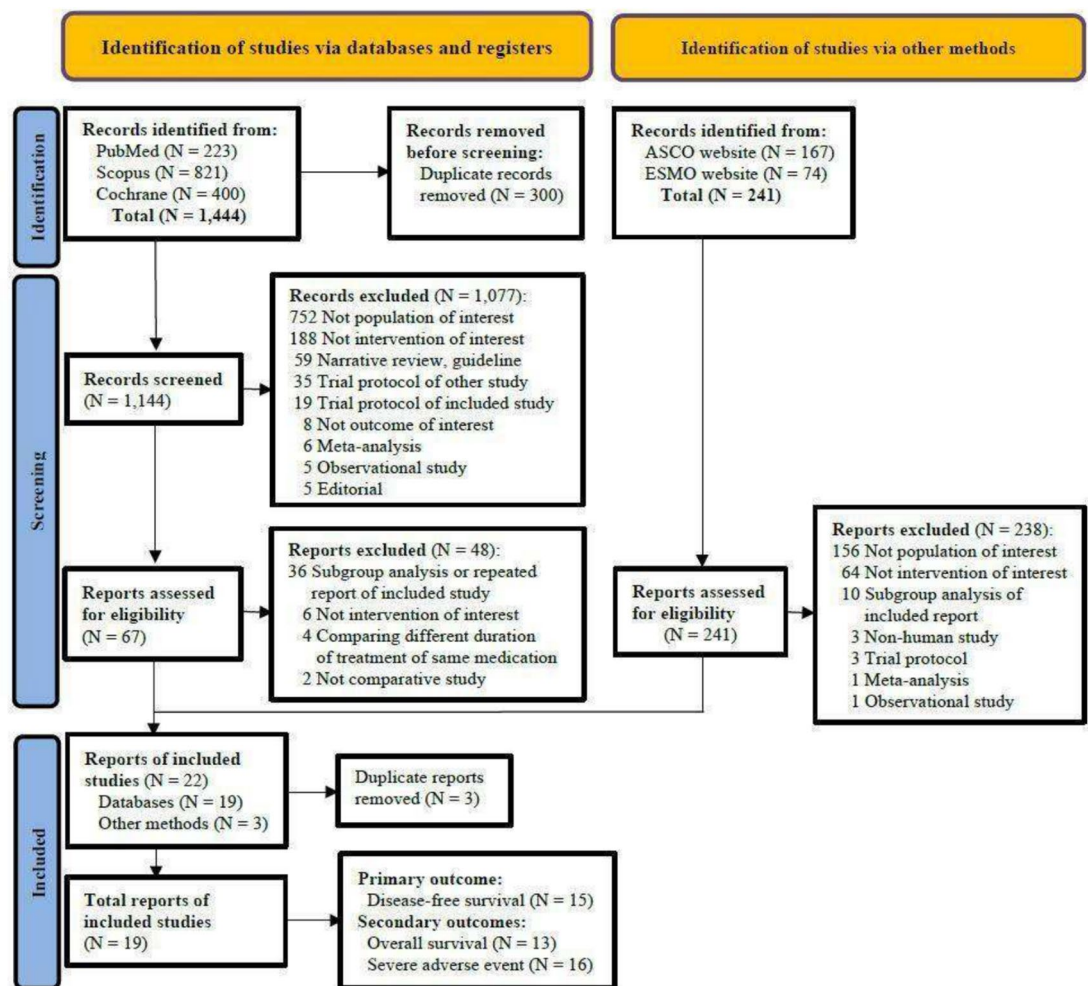


Fig. 1. Flow chart for study selection.

Of the 19 RCTs, 13^{6,35,36,38–40,43–49}, 3^{7,8,50}, 2^{37,41}, and 1⁴² compared EGFR-TKIs (i.e., gefitinib, erlotinib, icotinib, or osimertinib), ICI immunotherapy (i.e., atezolizumab or pembrolizumab), non-ICI immunotherapy (i.e., MAGE-A3 immunotherapeutic), and VEGF inhibitor (i.e., bevacizumab) with chemotherapy/placebo, respectively.

Risk of bias assessment

Of all included studies, 73.7% were classified as low risk of bias, while 26.3% had some concerns regarding the randomization process, deviations from intended interventions, and selection of the reported results; see supporting information (Supplementary Fig. S1).

Direct meta-analysis

A direct MA was performed on EGFR-TKIs vs. chemotherapy/placebo using data from 10 studies^{6,36,38–40,44–48} involving 3,263 patients. The estimated median DFS for patients treated with EGFR-TKIs was significantly longer compared to those who received chemotherapy/placebo, with a corresponding median DFS (95% CI) of 64.24 months (32.73, 95.75) and 36.69 months (23.20, 50.18), respectively. Furthermore, EGFR-TKIs significantly reduced disease recurrence, showing a HR of 0.53 (95% CI: 0.35, 0.80; p-value = 0.002) compared to chemotherapy/placebo, and there was evidence of high heterogeneity with an I^2 of 92.02% and a Q test chi-square value of 112.72 (p-value < 0.001); see supporting information in Supplementary Table S3 and Supplementary Fig. S2. The results of Egger's test (coef. = -2.00, p-value = 0.0864), along with the funnel plot and contour-enhanced funnel plot, showed no evidence of publication bias (Supplementary Fig. S3).

Meta-regression (Supplementary Table S4) was conducted to explore sources of heterogeneity. Of 10 included studies, eight were conducted in patients with EGFR mutation-positive, one was based on EGFR expression, and one did not report EGFR mutation status. A sensitivity analysis was performed by excluding the two studies with unclear or non-mutation-based EGFR status, focusing on only EGFR-mutation-positive studies (n = 1,787). The results demonstrated a consistent trend in favor of EGFR-TKI treatment, with a median DFS of 43.80 months (95% CI: 21.03, 66.57), compared to 22.32 months (95% CI: 14.85, 29.79) in the chemotherapy/placebo. The estimated HR was 0.43 (95% CI: 0.27, 0.68; p-value < 0.001), indicating significantly reduced risk of disease

Author, year	Intervention (n)	Comparator (n)	RCT Phase	Region	EGFR mutation status	Included staging
Tsuboi M, 2005 ³⁵	Gefitinib: TKI (n = 18)	Placebo (n = 20)	III	Japan	Not specified	IB—IIIA
Goss GD, 2013 ³⁶	Gefitinib: TKI (n = 251)	CMT / placebo (n = 252)	III	North America	Not specified	IB—IIIA
Vansteenkiste J, 2013 ³⁷	MAGE-A3: non-ICI (n = 122)	Placebo (n = 60)	II	Europe	Not specified	IB—II
Li N, 2014 ³⁸	Gefitinib: TKI (n = 30)	CMT / placebo (n = 30)	II	China	EGFR mutation	IIIA (N2)
Feng S, 2015 ³⁹	Icotinib: TKI (n = 21)	CMT / placebo (n = 18)	II	China	EGFR mutation	IB (high risk)—IIIA
Kelly K, 2015 ⁴⁰	Erlotinib: TKI (n = 623)	CMT / placebo (n = 350)	III	International	EGFR expression	IB—IIIA (microscopic N2 only)
Vansteenkiste JE, 2016 ⁴¹	MAGE-A3: non-ICI (n = 1,515)	CMT / placebo (n = 757)	III	International	Not specified	IB—IIIA
Wakelee HA, 2017 ⁴²	Bevacizumab: VEGF inh (n = 752)	CMT / placebo (n = 749)	III	International	Not specified	IB (≥ 4 cm)—IIIA
Zhong WZ, 2018 ⁴³	Gefitinib: TKI (n = 111)	CMT / placebo (n = 111)	III	China	EGFR mutation	II—IIIA (N1—N2)
Felip E, 2021 ⁷	Atezolizumab: ICI (n = 507)	CMT / placebo (n = 498)	III	International	Not specified	IB—IIIA
He J, 2021 ⁴⁴	Icotinib: TKI (n = 161)	CMT / placebo (n = 161)	III	China	EGFR mutation	II—IIIA
Zhong WZ, 2021 ⁴⁵	Gefitinib: TKI (n = 111)	CMT / placebo (n = 111)	III	China	EGFR mutation	II—IIIA (N1—N2)
O'Brien M, 2022 ⁸	Pembrolizumab: ICI (n = 590)	CMT / placebo (n = 587)	III	International	Not specified	IB—IIIA
Tada H, 2022 ⁴⁶	Gefitinib: TKI (n = 116)	CMT / placebo (n = 116)	III	Japan	EGFR mutation	II—III
Yue D, 2022 ⁴⁷	Erlotinib: TKI (n = 51)	CMT / placebo (n = 51)	II	China	EGFR mutation	IIIA
Herbst RS, 2023 ⁶	Osimertinib: TKI (n = 339)	CMT / placebo (n = 343)	III	International	EGFR mutation	IB—IIIA
Ou W, 2023 ⁴⁸	Icotinib: TKI (n = 63)	Placebo (n = 65)	II	China	EGFR mutation	IB
Tsuboi M, 2023 ⁴⁹	Osimertinib: TKI (n = 339)	CMT / placebo (n = 343)	III	International	EGFR mutation	IB—IIIA
Felip E, 2023 ⁵⁰	Atezolizumab: ICI (n = 507)	CMT / placebo (n = 498)	III	International	Not specified	IB—IIIA

Table 1. Characteristics of the included studies.

recurrence. However, substantial heterogeneity was remained ($I^2=88.37\%$; $Q=60.17$, $p\text{-value}<0.001$; see Supplementary Fig. S4–S5).

Eight studies^{36,38,40,44–47,49}, including 3,096 patients, compared OS between EGFR-TKIs and chemotherapy/placebo. The estimated median OS (95% CI) for patients treated with EGFR-TKIs was not significantly different from that of patients receiving chemotherapy/placebo, i.e., 107.78 months (75.02, 140.53) vs. 95.82 months (66.71, 124.94), respectively. The corresponding HR (95% CI) was 0.81 (0.62, 1.07; $p\text{-value}=0.136$), with evidence of high heterogeneity ($I^2=71.94\%$; $Q\text{ test: chi-square}=24.95$, $p\text{-value}=0.001$); see supporting information (Supplementary Table S5 and Supplementary Fig. S6). The results of Egger's test (coef. = -2.75, $p\text{-value}=0.0644$), along with the funnel plot and contour-enhanced funnel plot, showed no evidence of publication bias (Supplementary Fig. S7).

Eleven studies^{6,35,36,38–40,43,44,46–48}, involving 3,202 patients, evaluated SAE in comparing EGFR-TKIs and chemotherapy/placebo. There was no significant difference in the SAE rate with a pooled RR (95% CI) of 1.11 (0.57, 2.17; $p\text{-value}=0.759$), with evidence of high heterogeneity ($I^2=91.04\%$; $Q\text{ test: chi-square}=111.63$, $p\text{-value}<0.001$); see Supporting information (Supplementary Table S6 and Supplementary Fig. S8). The results of Egger's test (coef. = 2.36, $p\text{-value}=0.0058$) and the funnel plot indicated asymmetry; however, the contour-enhanced funnel plot showed no publication bias (Supplementary Fig. S9).

Heterogeneity was high for all outcomes, but no sources of heterogeneity were detected. The results are presented in Supplementary Table S4.

Network meta-analysis

Disease-free survival

Fifteen RCTs^{6–8,36–42,44–48}, involving 9,400 patients, were included in the NMA of five treatment regimens. The network map is shown in Supplementary Fig. S10a. A mixed-effect parametric regression with a log-normal survival distribution was selected, as per the smallest AIC (Supplementary Table S7), to estimate the median DFS.

The estimated median DFS for EGFR-TKIs and ICI immunotherapy was longer than that of chemotherapy/placebo, with median DFS (95% CI) of 69.24 months (44.52, 93.97), 53.47 months (-7.35, 114.28), and 39.49 months (29.32, 49.67), respectively; however, these differences were not statistically significant. Neither VEGF inhibitor nor non-ICI immunotherapy differed significantly relative to chemotherapy/placebo, with median DFS of 39.33 (-2.61, 81.26) and 39.63 (-21.01, 100.28) months (Fig. 2A). The HRs (95% CI) for DFS were estimated, and it was found that patients who received EGFR-TKIs, ICI immunotherapy, and non-ICI immunotherapy were 0.54 (0.38, 0.77; p-value=0.001), 0.78 (0.39, 1.56; p-value=0.475), and 0.95 (0.46, 1.95; p-value=0.885) times less likely to experience recurrence compared with chemotherapy/placebo, respectively; only the effect for EGFR-TKIs was significant. Conversely, patients receiving a VEGF inhibitor had essentially similar risk of recurrence relative to chemotherapy/placebo (HR=1.02 (0.38, 2.71)) (Table 2). The treatment ranking based on SUCRA values showed that EGFR-TKIs had the highest of the most effective (SUCRA 90.6%), followed by ICI immunotherapy (SUCRA 56.0%), non-ICI immunotherapy (SUCRA 37.5%), VEGF inhibitors (SUCRA 36.4%), and chemotherapy/placebo (SUCRA 29.5%). These results suggested that EGFR-TKIs might be the most effective option for reducing the risk of recurrence.

There was no evidence of inconsistency in the network (chi-square=0.02, p-value=0.885). The sensitivity analysis, conducted by removing studies with a some-concern risk of bias (Supplementary Table S8), showed that the DFS results and SUCRA ranking remained consistent with the overall study findings.

The comparison-adjusted funnel plot (Supplementary Fig. S11a) showed asymmetry, mainly driven by the EGFR-TKIs vs. chemotherapy/placebo comparison. A contour-enhanced funnel plot was subsequently constructed for this comparison (Supplementary Fig. S11b), which showed that studies symmetrically distributed across both significant and non-significant areas. This suggested that observed funnel plot asymmetry was more likely due to heterogeneity rather than missing studies or publication bias.

Overall survival

The data from 13 RCTs^{8,36–38,40–42,44–47,49,50}, involving 9,233 patients, assessed the effect of five treatment regimens on OS. The network map is shown in Supplementary Fig. S10b. A mixed-effect parametric regression with a log-normal survival distribution was selected to estimate the median OS based on the smallest AIC (Supplementary Table S7).

The estimated median OS (95% CI) for chemotherapy/placebo, EGFR-TKIs, VEGF inhibitors, ICI immunotherapy, and non-ICI immunotherapy was 103.24 (82.21, 124.26), 116.83 (91.36, 142.31), 103.81 (76.75, 130.87), 105.72 (79.49, 131.95), and 101.85 months (76.99, 126.72), respectively (Fig. 2B). Compared to chemotherapy/placebo, EGFR-TKIs, VEGF inhibitors, and ICI immunotherapy were associated with a reduction in the risk of death by approximately 18% [HR=0.82 (0.64, 1.05)], 1% [HR=0.99 (0.57, 1.72)], and 6% [HR=0.94 (0.62, 1.42)], respectively; however, these differences were not statistically significant. Additionally, non-ICI immunotherapy showed no significant difference in the risk of death [HR=1.00 (0.64, 1.55)] (Table

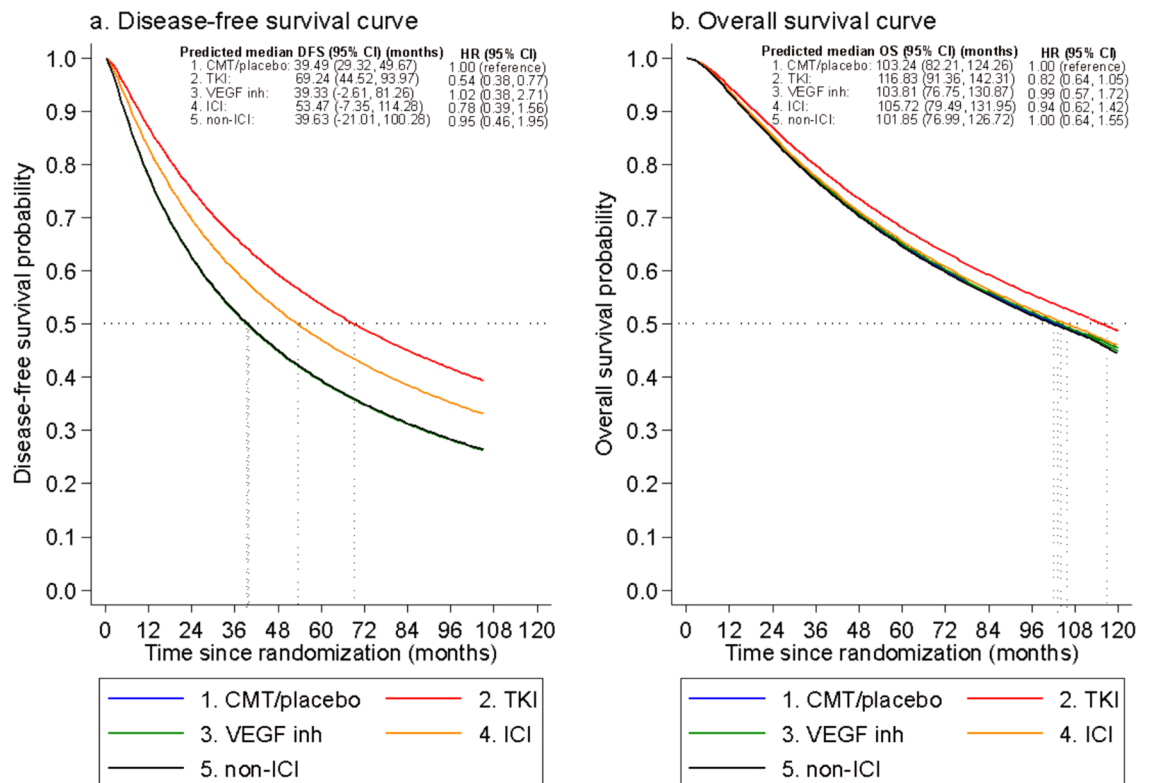


Fig. 2. Survival curve of DFS and OS outcome using multi-level mixed-effect parametric survival analysis.

Reference treatment	Hazard ratio (95% CI)				
	CMT/placebo	TKI	VEGF inh	ICI	Non-ICI
CMT/placebo	29.5, 0.0	0.54 (0.38, 0.77)	1.02 (0.38, 2.71)	0.78 (0.39, 1.56)	0.95 (0.46, 1.95)
TKI	1.84 (1.29, 2.63)	90.6, 68.3	1.88 (0.66, 5.31)	1.43 (0.65, 3.13)	1.75 (0.78, 3.90)
VEGF inh	0.98 (0.37, 2.61)	0.53 (0.19, 1.51)	36.4, 10.8	0.76 (0.23, 2.53)	0.93 (0.28, 3.14)
ICI	1.29 (0.64, 2.59)	0.70 (0.32, 1.53)	1.31 (0.39, 4.36)	56.0, 15.3	1.22 (0.45, 3.33)
Non-ICI	1.05 (0.51, 2.16)	0.57 (0.26, 1.28)	1.07 (0.32, 3.61)	0.82 (0.30, 2.23)	37.5, 5.6

Table 2. Estimated relative treatment effects by HRs on DFS. Each diagonal cell (Bold) contained a surface under the cumulative ranking curve (SUCRA) value (first value) and percentage probability of being the best treatment for the lowest risk of recurrence (second value). Each off-diagonal cell (Roman) contained HR (95% CI). Highlight (Italics) was the statistically significant comparison. HR, hazard ratio; CI, confidence interval; DFS, disease-free survival; CMT/placebo, chemotherapy/placebo; TKI, EGFR-TKIs; VEGF inh, VEGF inhibitor; ICI, ICI immunotherapy; non-ICI, non-ICI immunotherapy.

Reference treatment	Hazard ratio (95% CI)				
	CMT/placebo	TKI	VEGF inh	ICI	Non-ICI
CMT/placebo	35.4, 0.7	0.82 (0.64, 1.05)	0.99 (0.57, 1.72)	0.94 (0.62, 1.42)	1.00 (0.64, 1.55)
TKI	1.21 (0.95, 1.56)	80.3, 47.8	1.20 (0.66, 2.20)	1.15 (0.71, 1.85)	1.22 (0.73, 2.01)
VEGF inh	1.01 (0.58, 1.75)	0.83 (0.45, 1.52)	44.7, 20.5	0.95 (0.48, 1.89)	1.01 (0.50, 2.04)
ICI	1.06 (0.70, 1.60)	0.87 (0.54, 1.41)	1.05 (0.53, 2.09)	50.0, 17.8	1.06 (0.58, 1.94)
Non-ICI	1.00 (0.64, 1.55)	0.82 (0.50, 1.36)	0.99 (0.49, 2.00)	0.94 (0.52, 1.72)	39.7, 13.2

Table 3. Estimated relative treatment effects by HRs on OS. Each diagonal cell (Bold) contained a surface under the cumulative ranking curve (SUCRA) value (first value) and percentage probability of being the best treatment for the lowest risk of death (second value). Each off-diagonal cell (Roman) contained HR (95% CI). Highlight (Italics) was the statistically significant comparison. HR, hazard ratio; CI, confidence interval; OS, overall survival; CMT/placebo, chemotherapy/placebo; TKI, EGFR-TKIs; VEGF inh, VEGF inhibitor; ICI, ICI immunotherapy; non-ICI, non-ICI immunotherapy.

3). The ranking by SUCRA indicated that EGFR-TKIs had the highest probability of reducing the risk of death (SUCRA 80.3%), followed by ICI immunotherapy (SUCRA 50.0%). The consistency assumption was checked, suggesting no evidence of inconsistency (chi-square < 0.01, p-value = 0.998).

The comparison-adjusted funnel plot (Supplementary Fig. S12a) indicated asymmetry in comparing EGFR-TKIs vs. chemotherapy/placebo. However, the contour-enhanced funnel plot for this pairwise comparison (Supplementary Fig. S12b) showed that studies were distributed across both significant and non-significant regions. Therefore, the observed asymmetry is more likely attributable to heterogeneity rather than publication bias.

Severe adverse events

A total of 16 RCTs^{6,8,35–44,46–48,50}, involving 9,280 patients, were included in the NMA of SAE, considering five different treatment regimens. The network map is shown in Supplementary Fig. S10c. The common SAEs associated with different treatment regimens were as follows: abnormal liver function test, rash/acne, and diarrhea for EGFR-TKIs; hypertension for VEGF inhibitors; abnormal liver function test and hypertension for ICI immunotherapy; and myalgia/arthralgia and fatigue for non-ICI immunotherapy.

ICI immunotherapy, EGFR-TKIs, VEGF inhibitors, and non-ICI immunotherapy were more likely to develop SAE than chemotherapy or placebo, with risk ratios (95% CI) of 1.60 (0.29, 8.91), 1.24 (0.52, 2.91), 1.23 (0.11, 13.72), and 1.08 (0.19, 6.19), respectively, but these results were not statistically significant (Table 4). The treatment with the highest chance of serious side effects was ICI immunotherapy (SUCRA 63.9%), followed by EGFR-TKIs (SUCRA 52.6%) and VEGF inhibitors (SUCRA 49.4%), respectively. There was no evidence of inconsistency across the network (chi-square = 0.01, p-value = 0.930).

The comparison-adjusted funnel plot indicated asymmetry due to some studies of EGFR-TKIs vs. chemotherapy/placebo (Supplementary Fig. S13a). The contour-enhanced funnel plot of this comparison

Reference treatment	Risk ratio (95% CI)				
	CMT/placebo	TKI	VEGF inh	ICI	Non-ICI
CMT/placebo	37.6, 2.0	1.24 (0.52, 2.91)	1.23 (0.11, 13.72)	1.60 (0.29, 8.91)	1.08 (0.19, 6.19)
TKI	0.81 (0.34, 1.91)	52.6, 12.6	1.00 (0.08, 12.87)	1.30 (0.19, 8.83)	0.87 (0.13, 6.11)
VEGF inh	0.81 (0.07, 8.99)	1.00 (0.08, 12.90)	49.4, 29.9	1.30 (0.07, 25.01)	0.88 (0.04, 17.13)
ICI	0.62 (0.11, 3.43)	0.77 (0.11, 5.20)	0.77 (0.04, 14.69)	63.9, 36.6	0.67 (0.06, 7.73)
Non-ICI	0.92 (0.16, 5.30)	1.14 (0.16, 7.99)	1.14 (0.06, 22.35)	1.49 (0.13, 17.16)	46.5, 18.9

Table 4. Estimated relative treatment effects by RRs on SAE. Each diagonal cell (Bold) contained a surface under the cumulative ranking curve (SUCRA) value (first value) and percentage probability of being the worst treatment for the highest incidence of severe adverse event (second value). Each off-diagonal cell (Roman) contained RR (95% CI). Highlight (Italics) was the statistically significant comparison. RR, risk ratio; CI, confidence interval; SAE, severe adverse event; CMT/placebo, chemotherapy/placebo; TKI, EGFR-TKIs; VEGF inh, VEGF inhibitor; ICI, ICI immunotherapy; non-ICI, non-ICI immunotherapy.

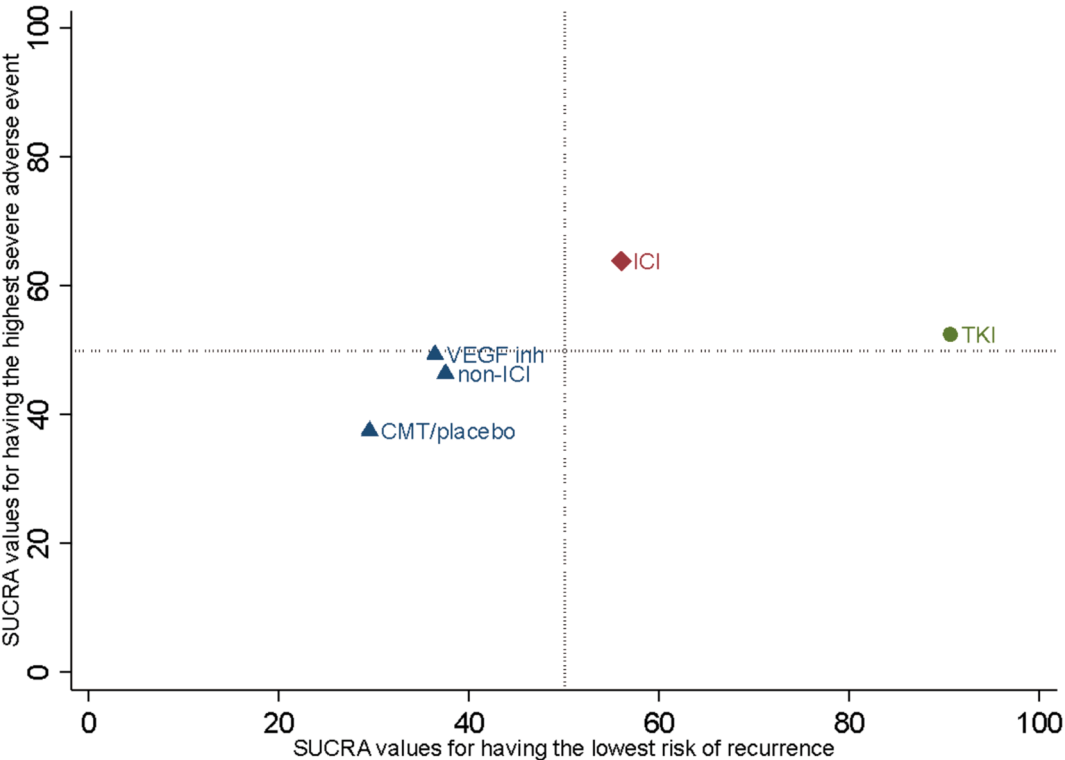


Fig. 3. Clustered ranking plot based on SUCRA values for the lowest risk of recurrence versus the highest SAE.

revealed no evidence of missing studies (Supplementary Fig. S13b), suggesting that the funnel plot asymmetry was likely due to heterogeneity rather than publication bias.

Cluster ranking analysis

A clustered ranking plot was constructed to visualize the ranks of both benefit (DFS) and risk (SAE); see Fig. 3. This graph plots the probability for a lower recurrence risk on the x-axis and a higher SAE on the y-axis. Treatment regimens in the lower right corner show higher benefits but lower risks, while those in the upper left corner exhibit the opposite.

The clustered ranking plot demonstrated three clusters representing the five treatment regimens based on similarity in their benefit and risk outcomes. EGFR-TKIs emerged as the best treatment option among these regimens, with the highest benefit (SUCRA 90.6%) and intermediate SAE risk (SUCRA 52.6%). The second-best

cluster was ICI immunotherapy, which had an intermediate benefit (SUCRA 56.0%) of reducing recurrence risk but the highest SAE risk (SUCRA 63.9%).

Evaluation of the certainty of evidence

The CINeMA framework was used to evaluate the certainty of evidence on DFS, OS, and SAE outcomes. Confidence ratings for the DFS outcome were generally low due to concerns about imprecision. The significant pairwise comparison of the DFS outcome between EGFR-TKIs and chemotherapy/placebo also had a low rating due to heterogeneity, as detailed in supporting information (Supplementary Table S9). For the OS outcome, the majority of confidence ratings were low, mainly due to imprecision (Supplementary Table S10). Likewise, most confidence ratings for the SAE outcomes were low, mainly due to substantial concerns related to imprecision (Supplementary Table S11).

Discussion

This study included 19 RCTs. An SRNMA looks at how different adjuvant treatment regimens (i.e., chemotherapy/placebo, EGFR-TKIs, VEGF inhibitors, ICI immunotherapy, and non-ICI immunotherapy) affect DFS, OS, and safety in patients with resected early-stage NSCLC. The main findings suggest that EGFR-TKIs might be the most effective option in reducing the recurrence risk; however, the certainty of the evidence is low. There is no strong evidence that EGFR-TKIs improve OS or increase the risk of SAE compared to other treatment regimens.

To our knowledge, this is the first SRNMA to simultaneously evaluate the efficacy and safety of all available adjuvant treatment regimens, including targeted therapy and immunotherapy, relative to chemotherapy for patients with resected early-stage NSCLC. The use of NMA allowed for indirect comparisons where no head-to-head trials were available, and the ranking probability of the best treatment may help clinicians assess the relative benefits of different options. EGFR-TKIs were the most effective regimen for prolonging DFS and OS, consistent with the findings of the previous SRMA¹³. However, the OS benefit was not statistically significant compared to chemotherapy/placebo.

Our findings demonstrated that ICI immunotherapy may represent a recommended second-line agent, improving DFS and OS and aligning with the results of the individual studies^{7,8} included in this NMA, although these differences were insignificant. This limitation may be attributed to the small number of studies included in our analysis. Compared with chemotherapy/placebo, non-ICI immunotherapy and VEGF inhibitors showed no improvement in DFS and OS. Consequently, the best treatment regimen based on the SUCRA ranking for lowering the risk of recurrence was EGFR-TKIs, followed by ICI immunotherapy, as observed in our NMA.

In the subgroup analysis based on EGFR mutation status, EGFR-TKIs significantly improved DFS in EGFR mutation-positive patients compared to chemotherapy/placebo, despite considerable heterogeneity. These findings suggest that EGFR-TKIs are highly effective in patients with EGFR mutations. In contrast, conclusions for the EGFR mutation-negative subgroup cannot be drawn due to the limited number of studies available.

Regarding safety outcomes, ICI immunotherapy, EGFR-TKIs, and VEGF inhibitors appeared to have higher SAE rates of 60%, 24%, and 23%, respectively, compared to chemotherapy/placebo. However, these differences did not reach statistical significance. These estimates were based on a few included studies and were imprecise due to the paucity of reported SAEs, meaning a more detailed safety analysis was not feasible due to inconsistent SAE reporting. Future studies using real-world data may better capture long-term and population-specific toxicity. An updated NMA will be required when additional RCTs are published. Therefore, the recommended use of EGFR-TKIs should be considered cautiously, especially in elderly patients. A previous study⁵¹ reported more severe toxicity and a higher rate of dose reduction in elderly patients prescribed EGFR-TKIs compared with younger patients. Nevertheless, health-related quality of life (HRQoL) in EGFR-TKIs was maintained during treatment without clinically meaningful differences compared with chemotherapy/placebo⁵². This evidence might support the use of adjuvant EGFR-TKIs to delay disease recurrence without compromising HRQoL. Further studies should be conducted to explore the HRQoL benefit of adjuvant treatment.

Although EGFR-TKIs rank highest and are recommended for reducing disease recurrence risk by both the European Society for Medical Oncology (ESMO) and the American Society of Clinical Oncology (ASCO)^{53,54}, particularly Osimertinib, their high cost remains a significant concern, with an annual price of \$204,000 USD⁵⁵. As a result, affordability and accessibility are significant challenges, especially in low- to middle-income countries. Therefore, a cost-utility analysis is warranted before widespread clinical recommendations for this treatment regimen.

Our study had several notable strengths. No previous NMA has comprehensively examined all available adjuvant treatment regimens and determined the most appropriate treatment approach for resected early-stage NSCLC patients. Thus, our NMA is the first attempt to summarize objective evidence of the treatment efficacy and safety of targeted therapy and immunotherapy. Our study generated individual patient data based on Kaplan–Meier curves of DFS and OS. This analytical method provided greater flexibility in modeling by utilizing a mixed-effect survival model^{27,28}. Consequently, the median time for DFS and OS was obtained. This median time information is valuable for patient counseling, enabling clinicians to illustrate the benefits associated with each adjuvant treatment regimen. Moreover, our study incorporated cluster ranking analysis to evaluate the risk–benefit profiles of each treatment regimen simultaneously. By considering both factors, medical oncologists can make better-informed decisions when selecting the most appropriate adjuvant treatment regimen that maximizes DFS while minimizing the occurrence of SAE.

Several limitations of our study should be acknowledged. First, only one study focused on Stage I NSCLC patients, while the remaining studies included mixed populations from states IB–IIIA. A subgroup analysis for Stage I NSCLC patients could not be performed. Additionally, not all studies included in our analysis focused on patients with EGFR mutation-positive. Nevertheless, a sensitivity analysis including only EGFR mutation-positive patients showed consistent effect of EGFR-TKI, although heterogeneity was remained. Second, the

number of studies available for direct pairwise comparisons was limited, aside from those comparing EGFR-TKIs with chemotherapy/placebo. This limitation reduced the precision of both treatment effect estimates and SUCRA-based treatment ranking, particularly for comparisons involving ICI immunotherapy versus chemotherapy/placebo, and EGFR-TKIs and ICI immunotherapy. Currently, there are many ongoing clinical studies of adjuvant ICI immunotherapy compared with chemotherapy/placebo for resected early-stage NSCLC patients^{56–58}. To generate more robust and precise results, it is crucial to regularly update NMA that includes data from these additional adjuvant immunotherapy studies as it becomes available. Furthermore, the pooled estimates showed high heterogeneity, and funnel plots were asymmetric, either reflecting this heterogeneity or, alternatively, indicating potential publication bias. This result suggests that despite our findings incorporating the most recently published data, the precision of our findings is limited by the available data, particularly for new treatment regimens, and they require further independent validation.

Conclusion

The findings of this NMA provide objective evidence suggesting that adjuvant targeted therapy or immunotherapy is more effective than chemotherapy/placebo in prolonging DFS. However, there is no strong evidence supporting improved OS in patients with early-stage NSCLC. EGFR-TKIs appear to be the most effective option for reducing the risk of recurrence and death, but SAEs should be closely monitored after prescription. Due to the low certainty of evidence, further updates to the NMA are needed to incorporate ongoing RCTs of adjuvant immunotherapy, which may provide more robust and precise estimates.

Data availability

Data is provided within supplementary information files.

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Author contributions

S.S., S.R., and A.T.- Conceptualization, Methodology, Formal analysis, Writing—original draft preparation S.S., P.P., and P.N.—Data curation S.S., S.R., P.P., and P.N.—Writing—review A.T., G. M., and J.A.— Writing—review and editing, Supervision. All authors reviewed the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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